Notes on Method of Undetermined Coefficients

from Pg 164, Spiegel "Applied Differential Equations" for solving a linear differential equation with constant coefficients.

1. Find the complementary solution y_c, the general solution of eq. (2) in our example.

$$\ddot{y} + y = 0 \qquad (2)$$

Here y is a function whose second derivative is the negative of y. This is true of $\sin(t)$ and $\cos(t)$, so the general solution of (2) is

$$y_c = c_1 \sin(t) + c_2 \cos(t) \qquad (3)$$

where c_1 and c_2 are arbitrary constants.

2. Assume a particular solution corresponding to the terms on the right side of the equation:

$$\ddot{y} + y = 3 + 3\cos(2t)$$

For terms $\sin(rt)$, $\cos(rt)$, assume $a\sin(rt) + b\cos(rt)$

So here we assumed

$$y_p(t) = A + B\cos(2t) + C\sin(2t)$$

Spiegel provides further rules for cases where polynomials and/or exponentials occur on the right side of the equation, and corrections for when assumed terms also occur in the complementary solution.